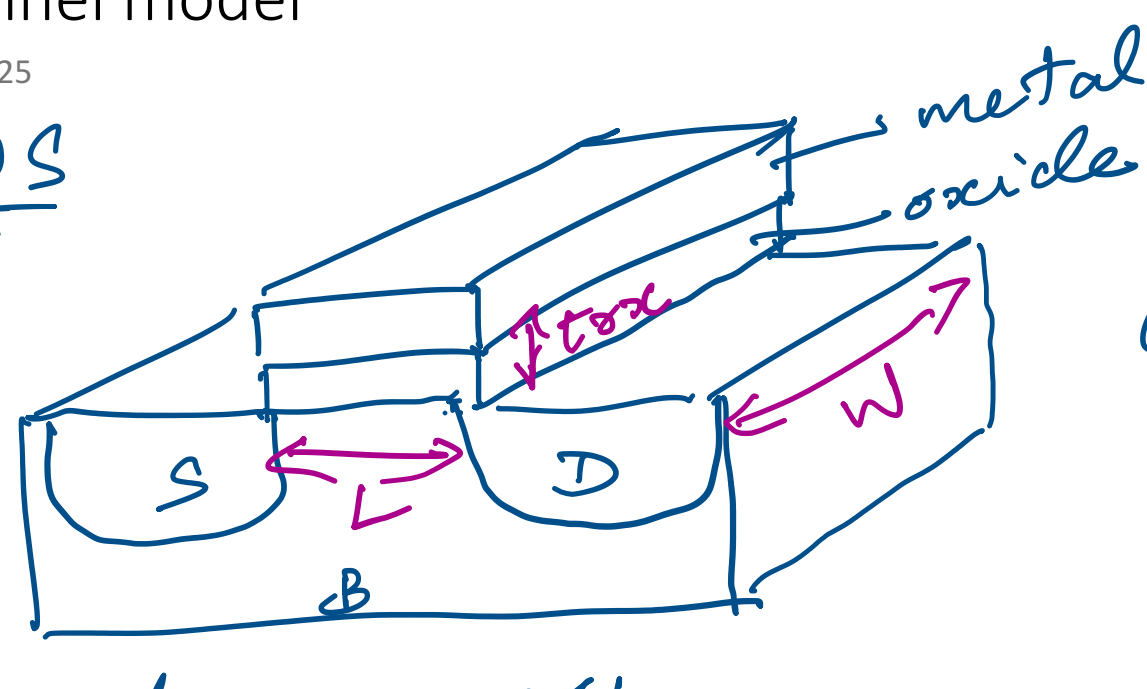


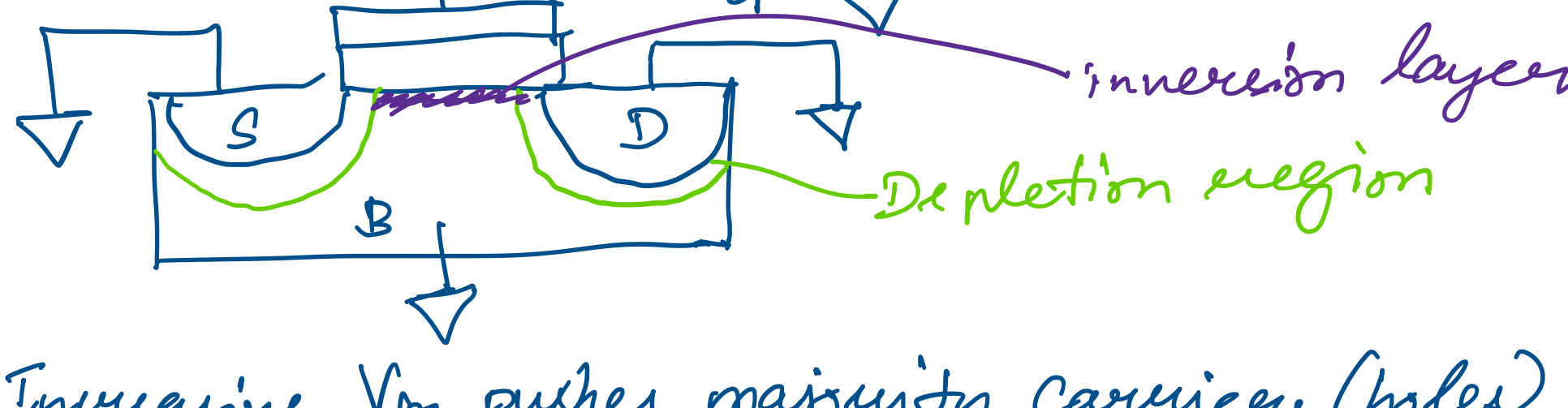
nMOS



$V_s \equiv$ reference voltage

$$I_{DS} = f(V_{GS}, V_{DS}, V_{BS})$$

Terminal	type
S, D	n^+
B	P
G	metal or polySi



Increasing V_{GS} pushes majority carriers (holes) in the p-substrate away from the oxide boundary. Increasing V_{GS} further starts attracting the minority carriers to the boundary - this is the formation of inversion layer.

The model we'll be using is that of a parallel plate capacitor between the gate and the bulk with the oxide layer forming the dielectric.

V_{TN} is the "magic" voltage at which the channel is inverted. $\Rightarrow$ For $V_{GS} < V_{TN}$ there is no channel. In reality there is no such magical voltage and there is a "weak" inversion where there are very few minority carriers $\Rightarrow$ tiny possible I_{DS} .

The capacitance C between the gate and body: $C = \frac{A \epsilon_{ox}}{t_{ox}}$

$$C_{ox} = \text{capacitance per unit area} = \frac{\epsilon_{ox}}{t_{ox}}$$

If we apply a voltage $V_{DS} > 0$, there is a current flow by the inversion charges I_{DS} .



Assuming a linear voltage drop across the channel $V(x)$ is the voltage at the distance x from the source.

Let $Q_I(x)$ be the charge per unit area at x :

$$Q_I(x) = \text{Cap. per unit area} \times \text{Volt. diff.} = C_{ox}(V_{GS} - V_{TN} - V(x))$$

$(V_{GS} - V_{TN})$ term appears since V_{TN} is assumed to be spent in channel inversion

Velocity of the carrier v_n in a bulk medium is given by $v_n = \mu_n E = \mu_n \frac{dV(x)}{dx}$

$$\text{Current at } x = Q_I \times A$$

$$= \frac{dx}{v} \left[\text{time to travel } dx \right]$$

$$= \frac{C_{ox}(V_{GS} - V_{TN} - V(x)) \times W \times dx}{dx/v_n}$$

$$I_{DS} = C_{ox}(V_{GS} - V_{TN} - V(x)) W \mu_n \frac{dv}{dx}$$

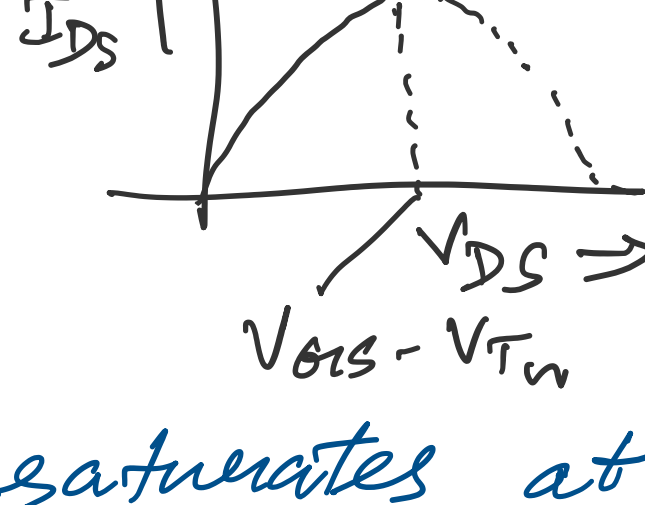
$$I_{DS} dx = C_{ox}(V_{GS} - V_{TN} - V(x)) W \mu_n dv$$

Integrating over whole length of channel

$$I_{DS} \int_0^L dx = \mu_n C_{ox} W \int_0^{V_{DS}} (V_{GS} - V_{TN} - V) dV$$

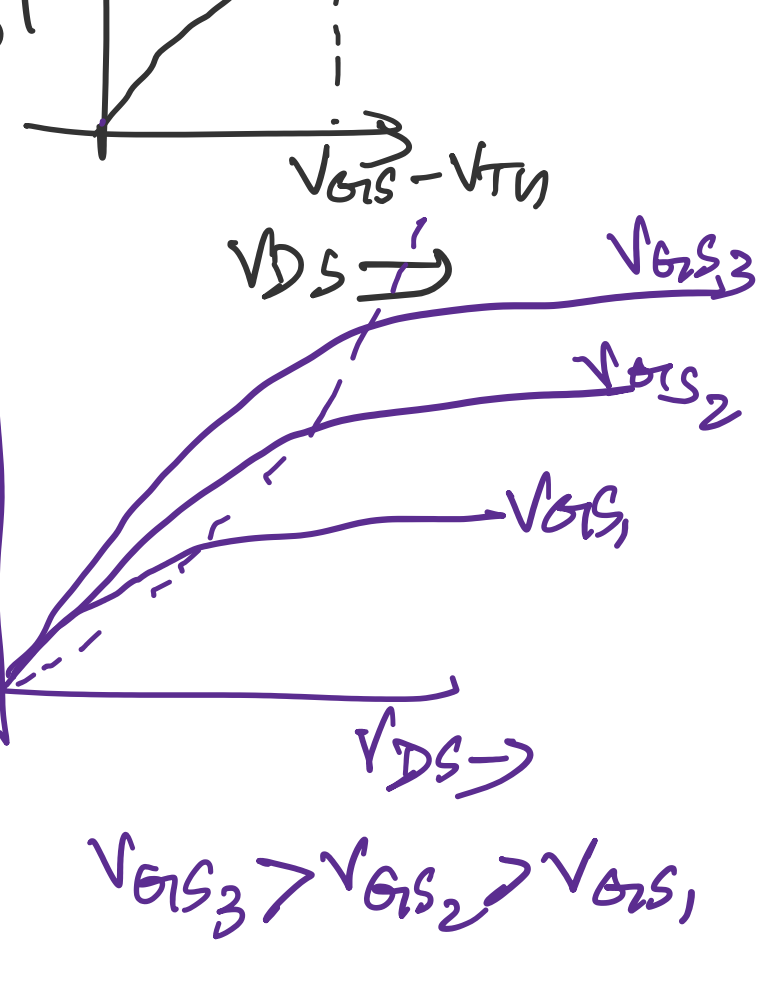
$$I_{DS} = \frac{\mu_n C_{ox} (W/L)}{2} \left[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2 \right]$$

Plot looks like this



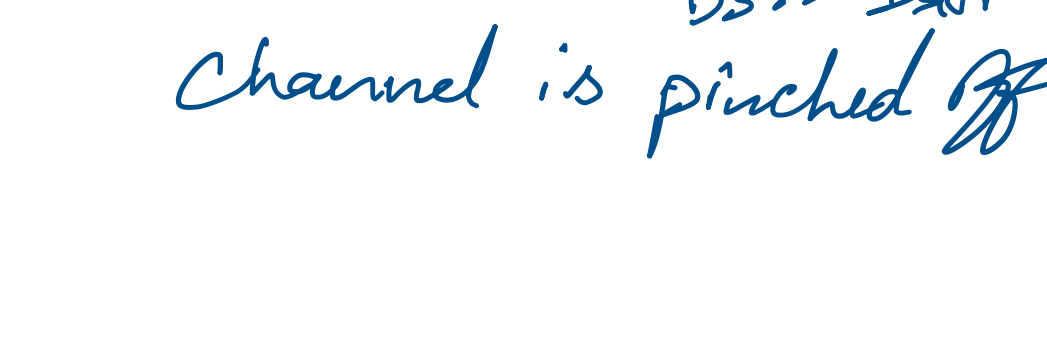
In reality current saturates at $V_{DS} = V_{GS} - V_{TN}$

$$I_{DSAT} = \frac{\mu_n C_{ox} (W/L)}{2} (V_{GS} - V_{TN})^2$$



Pinch off saturation and channel length modulation.

I_{DSAT} has a linear dependence on V_{DS} after saturation since the effective channel length reduces.



$$I_{DSAT} = \frac{\mu_n C_{ox} W}{2} (V_{GS} - V_{TN})^2 (1 + \lambda V_{DS})$$

channel length modulation parameter



Level-1 MOS model (nMOS)

$$I_D = 0 \quad \text{Cut-off} \quad V_{GS} < V_{TN}$$

$$= \frac{\mu_n C_{ox} (W/L)}{2} \left[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2 \right] (1 + \lambda V_{DS}) \quad \left| \begin{array}{l} V_{GS} \geq V_{TN} \\ V_{DS} < V_{GS} - V_{TN} \end{array} \right.$$

$$= \frac{\mu_n C_{ox} (W/L)}{2} (V_{GS} - V_{TN})^2 (1 + \lambda V_{DS}) \quad \left| \begin{array}{l} V_{GS} \geq V_{TN} \\ V_{DS} \geq V_{GS} - V_{TN} \end{array} \right.$$

Gradual Channel Approximation

pMOS p-channel MOSFET

$$I_D = 0, \quad V_{SG} < |V_{TP}| \quad \text{Cut-off}$$

$$I_n = \frac{\mu_p C_{ox} (W/L)}{2} \left[2(V_{SG} - |V_{TP}|)V_{SD} - V_{SD}^2 \right] (1 + \lambda V_{SD}) \quad \left| \begin{array}{l} V_{SG} \geq |V_{TP}| \\ V_{SD} < V_{SG} - |V_{TP}| \end{array} \right.$$

$$I_{sat} = \frac{\mu_p C_{ox} (W/L)}{2} (V_{SG} - |V_{TP}|)^2 (1 + \lambda V_{SD}) \quad \left| \begin{array}{l} V_{SG} \geq |V_{TP}| \\ V_{SD} \geq V_{SG} - |V_{TP}| \end{array} \right.$$

Scaling: the force behind Moore's law

Scaling Parameter ($s > 1$)

Parameter	Full Scaling	Constant Voltage Scaling
L'	L/s	L/s
W'	W/s	W/s
t_{ox}'	t_{ox}/s	t_{ox}/s
V_{DD}'	V_{DD}/s	V_{DD}
V_T'	V_T/s	V_T
I_{DSAT}'	I_{DSAT}/s	$I_{DSAT} \times s$
Power' = $V_{DD}' I_{DSAT}'$	Power/ s^2	Power $\times s$
Power density' = $\frac{\text{Power}'}{\text{Area}'}$	Power/Area	$s^3 \times \text{Power/Area}$

$\Downarrow$
Dennard scaling